Competition and Market Structure of NASDAQ

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Abstract

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1. Introduction

An essential feature of a dealer market like NASDAQ is competition for order flow among market makers. The competition presumably helps to ensure that investors get the best possible quotes and trade prices. Whether or not NASDAQ actually obtains competitive pricing is an empirical question. One view on this issue is that NASDAQ operates as a competitive market given a large number of competing market makers and fluid entry and exit. Wahal (1997), for instance, finds that entry and exit affect NASDAQ bid-ask spreads even after controlling for usual explanatory variables. Based on this finding, Wahal concludes that NASDAQ operates as a competitive market. However, other recent studies such as Ellis, Michaely and O'Hara (2000, 2002), and Schultz (2002) suggest that dominant NASDAQ market makers appear to exercise market power in setting bid-ask prices. Their finding implies that trading costs are higher than those in a competitive market, and that NASDAQ dealers benefit from higher trading costs.

In view of the far-reaching implications of their result, we feel it is important to take a deeper look at the relation between market structure and trading costs, and competition. In this paper, we examine the extent and dynamics of market power of NASDAQ market makers. In particular, we focus on two questions: (1) Do NASDAQ market makers enjoy statistically and economically significant market power in setting bid-ask prices? (2) If yes, then how persistent is the market power? We first establish the presence (or absence) of market power based on extensive dealer trading volume data over almost an eight-year period ending December 2000. Previous studies are based on selected sample over a few months. Focusing on a short period might obscure trends in economic rents especially if they tend to dissipate over longer time intervals. We next quantify the economic significance of rent that dominant market makers extract. Previous studies only look at statistical significance of rent, and have generally ignored the economic significance of dealer rents. Finally, we estimate the persistence of dealer rents. While the magnitude of rent is of concern to both investors and regulators, its persistence is also an importance issue. If competition works its way to reduce the rent rapidly, then market power is merely a transitory event. In this case,

there would be little need for policy intervention, even when rent is economically significant.

Our data include monthly share trading volume by every market maker for every stock traded on the NASDAQ National Market (NNM) during the period May 1995 to December 2000. This data set allows us to have an in-depth look at the time series patterns in competition among market makers. We find that concentration in trading affects bid-ask spreads even after controlling for the usual variables that explain bid-ask spreads. Our measure of market power or "rent" is statistically and economically significant: rent amounts to 8.76 cents or 0.54% of stock price. We classify stocks by dollar trading volume: the top 200 stocks (in trading volume) are classified as $-trading volume group 3, the next 800 stocks as group 2, the rest as group 1. The average market maker rent for those groups are approximately identical. What is different among these groups is the fraction of rent in stock price. Since stock prices and dollar trading volume are positively correlated, actively traded stocks tend to have a lower percentage of rent in price. The average percentage rent is 0.22% for group 3, 0.32% for group 2 and 0.58% for group 1. Half-life of rent is 20.3 months for entire sample, and 14.4 months for those stocks which undertook initial public offering during 1995-1997. Average half-lives of stocks in dollar trading volume groups are approximately equal.

This paper is closely related to a growing literature examining the behavior of dealers in dealer markets. Using extensive entry and exit data for 5,569 NASDAQ National Market stocks during a period from 1982 to 1993, Wahal (1997) finds that entry and exit affect bid-ask spreads even after controlling for usual explanatory variables. This effect is present even among stocks with a large number of market makers: entry is associated with a 6 cents decrease in end-of-day quoted spread for stocks with 2-7 market makers, and 2 cents decrease for stocks with more than 20 market makers. Based on this finding, Wahal concludes that NASDAQ operates as a competitive market.

Ellis, Michaely and O'Hara (2002) analyze market making activities for the first seven months for 313 NASDAQ stocks, which went public during September 1996 and July 1997. They find that the market for market making is highly concentrated with top

three dealers handling approximately 75% of trading volume. They observe positive relation between concentration of trading and bid-ask spreads: the more trading is concentrated, the higher the spread, given that all other things are constant. Another interesting finding is that market quickly reaches to a permanent equilibrium: "*By virtually every measure of market structure, the market that exists after the first three months of trading is the market that will characterize trading from then on*." We analyze dealers' behavior as firms reach their adolescent and more mature stages, and thus complement Ellis, Michaely and O'Hara (2002), which focus on very young firms during the first 7 months following their initial public offerings.

Klock and McCormick (1999), among others, document impact upon bid-ask spreads of the number of market makers and degree of concentration of trading among market makers. Klock and McCormick (1999) use Herfindahl index as measure of concentration in market making, and examine relation between Herfindahl index and bid-ask spreads for NASDAQ stocks for December 1996. Using dealer trading volume data for all NASDAQ stocks (on both NNM and Small Stock Market) for a period from May 1995 to February 1998, Schultz (2002) finds that dealers make markets only in stocks in which they have competitive advantage in accessing order flow and in information. He notes that those dealers with competitive advantages may appear to earn abnormally large returns, at least in the short-run. Our results are consistent with Schultz' intuition. Market makers indeed appear to earn non-trivial rent with half-life of 14.4 months for IPO stocks.

Using data for 180 days of trading surrounding implementation of new rules for 88 NASDAQ stocks, Weston (2000) examines competitive structure of the NASDAQ in the wake of new order handling rules in early 1997. Comparing components of bid-ask spreads, he finds that order processing components and market maker rent over all trades have fallen by 10 cents per trade after the new rules. The time-series of our estimate of market maker rent also show a decline of rent in early 1997 by approximately 4 cents for entire sample. Several studies examine interdealer trading (Reiss & Werner (1998)), preferencing, dealer profits and inventory (Hansch, Naik and Viswanathan, (1998, 1999)) based on data from the London Stock Exchange.

This paper is organized as follows. In section 2, we describe the data and key variables we use for our analysis. In section 3, we look into relation between concentration of trading among market makers and bid-ask spreads. We show that the impact of concentration on spreads are significant both in statistical and economic sense. In section 4, we analyze time series properties of the relation between concentration and spreads. Section 5 contains conclusion of the paper.

2. Data

The data contain monthly observations for all NASDAQ stocks during a period from May 1995 to December 2000. NASDAQ provided monthly share trading volume data of all dealers for each of stocks they are making markets in. We obtain trade and quote data data from the NYSE's TAQ database. To avoid problems with low trading volume, we exclude all stocks on the SmallCap market. We also exclude, from stocks on the NASDAQ National Market (NNM), stocks with average closing price below $2, and with average daily dollar trading volume below $10,000. The number of stocks each month in the sample ranges from 3,180 to 3,623.

Table 1 contains descriptive statistics of the sample stocks. Panel A shows that the distribution of daily dollar trading volume is heavily skewed to the right. Over entire sample period, the mean daily dollar trading volume is $9,163,000, while the median is only $475,500, which is just over 5% of the mean. Daily dollar trading volume has increased by approximately 7-fold, from $3.29 million in 1995 to $23.4 million in 2000.

The average trade size, however, has declined from $27,382 in 1995 to $15,358 in 2000. The mean of trade size over entire sample is $22,709 and the median is $15,185. A big increase in daily dollar trading volume and a decline in average trade size suggests that there has been a huge increase in number of trades. Panel C shows that it is indeed the case. Daily number of trades has gone up close to 10 times, from 71.8 in 1995 to 702.6 in 2000. The increase in daily dollar trading volume is due to more trades rather than bigger trades. Schultz (2002) has also observed a similar pattern in his sample of entire NASDAQ stocks over a period from May 1995 to February 1998.

Average daily closing price during sample period has increased from $16.65 in 1995 to $21.01 in 2002. Average closing price during entire period is $17.92, while median is $13.09. We use effective spreads and percentage effective spreads as the measure of trading cost. Effective spreads of a trade at time *t* of stock *i* are calculated as

$$\text{Effective Spread}_{it} = 2 * \mid P_{it} - (\text{Bid}_{it} + \text{Ask}_{it})/2 \mid \qquad (1)$$

where P_{it} is the trade price at time *t*, Bid_{it} and Ask_{it} are most recently posted inside bid and ask prices of stock *i*. We estimate percentage effective spreads by dividing the effective spreads by the midpoint of the contemporaneous inside bid and ask prices. Following Lee and Ready (1991), we use inside bid and ask prices at least five seconds old. Panels E and F show effective spreads and percentage effective spreads. Both have steadily declined over the sample period: the mean spread in 2000 is $0.2516 compared to $0.3822 in 1995, and the mean percentage spread is 2% in 2000 compared to 3.19% in 1995. The mean and median spreads over sample period are $0.3075 and $0.236, while the mean and median percentage spreads are 2.45% and 1.97%.

Last two variables in Table 1 are measures of competition among market makers. Panel G shows number of active market makers. Ellis et al (2002) observe that almost one third of all market makers do not trade even once a week. Accordingly, we count number of *active* market makers, defined as those with more than 0.1% share in total trading volume for a stock in a given month. Overall, the number of active market makers has increased from 11.6 in 1995 to 16.5 in 2000. The mean and median numbers of market makers are 13.2 and 11 over entire sample period. Previous studies (See Stoll (1978) and Wahal (1997) for instance) used the number of market makers as a measure of competition. In this paper, we use the Herfindhal-Hirschman index (HHI) as the measure of competition. The HHI for stock *i* during month *t* is defined as

$$\text{HHI}_{i,t} = \Sigma_{j = 1, N} \ (S_{j,t})^2 \qquad (2)$$

where $S_{j,t}$ is the percentage of stock *i*'s trading volume handled by market maker *j* during month t, and N denotes total number of market makers for stock *i*. For instance,

if four market makers handle 25% each of stock *i*'s trading volume, then the HHI = $(25)^2 + (25)^2 + (25)^2 + (25)^2 = 2{,}500$. It is easy to see that there are two components in the HHI:

$$\text{HHI} = 10{,}000 * [\ 1/N + N*(\text{Variance of market makers' market share})\ ] \qquad (3)$$

According to equation (3), two factors affecting the HHI are the number of market makers and how market shares are distributed among market makers, as measured by the variance of market shares. If there are four market makers each handling 25% of trading volume, then variance in eq. (3) is equal to zero, and HHI is equal to 2,500. When distribution of market shares deviates from the equal market share case, variance will become non-zero. As a result, the HHI will be higher than 2,500. The HHI is widely used as a measure of competition or concentration at an industry level both by researcher and regulatory body. For instance, the Federal Trade Commission uses a guideline that a HHI below 1,000 indicates no concentration, HHI between 1,000-1,800 moderate concentration and HHI over 1,800 severe concentration. The guideline of the Federal Trade Commission should be helpful in interpreting the HHI as applied to the NASDAQ market making.

Panel H shows the HHI. HHI has steadily decreased during sample period from 2,261 in 1995 to 1,748 in 2000. (See also Figure 1A.) The decline of HHI appears to occur over a broad range of NNM stocks. Each month, we classify stocks into three $-volume groups according to their dollar trading volume. Top 200 stocks in terms of $-trading volume are classified as $-volume group 3, next 800 stocks as $-volume group 2 and the rest $-volume group 1. As Figure 1B shows, HHI has declined for all these three $-volume groups. Note also that the HHI is skewed to the right: The mean of HHI is 2,133 while the median is 1,734. Lastly, as Schultz (2002) points out, the number of market makers relative to the HHI overstates the degree of competition: the mean HHI of 2,133 roughly corresponds to having five market makers with equal share, while the mean number of market makers is 13.2. Table 2 shows correlation among key variables.

The evidence is that HHI has declined on NASDAQ. Does it imply that NASDAQ has become more competitive over time, and that dominant market makers wield less market power in setting the bid and ask prices? This is the question we now turn to.

3. Concentration and Trading Costs

Competition among market makers is the central feature of a dealer market such as the NASDAQ. Indeed, the following statement of the NASDAQ fully recognizes the importance of competition:

> *"Key to NASDAQ's market structure are a core group of financial firms called market makers. ... The aggressive competition for orders fostered among NASDAQ's market participants helps to ensure that investors receive the best prices for the securities they trade."*

In a competitive market, market makers would be able to wield only minimal market power in setting the bid and ask prices. Even when they do exercise significant market power and thus set wide spreads, it should not be persistent. Wide spreads will attract new entries and foster more competition to bring the spreads down. Therefore, a measure of concentration would not affect trading costs in a competitive market. In order to examine the effect of concentration on trading costs, we run the following cross-sectional regression each month of the sample period,

$$\% \text{ Spread}_i = \alpha_0 + \alpha_1 \text{HHI}_i + \alpha_2 \ln(\text{TSize}_i) + \alpha_3 \ln(\text{Ntrades}_i) + \alpha_4 \sigma^2_i + \alpha_5 [1/E(P_i)] + \alpha_6 N_i + \varepsilon_i \quad (4)$$

where *i* denotes stock *i*, % Spread_i the mean of percentage effective spread, HHI_i is the Herfindahl index for the month, $\ln(\text{TSize}_i)$ is the natural logarithm of the mean trade size in dollar, $\ln(\text{Ntrades}_i)$ the natural logarithm of the mean number of trades, σ^2_i variance of stock prices, $E(P_i)$ the mean closing price, and N_i the number of market makers for stock *i*. We use the following estimate of the variance of stock prices:

$$\hat{\sigma}^2_{i,t} \equiv \frac{\sum_{q=1}^{Q}\left(\frac{B_q + A_q}{2}\right)^2 - (\text{Average of bid} - \text{ask midpoints for the day})^2}{(\text{Average of bid} - \text{ask midpoints for the day})^2} \quad (5)$$

where Q is the number of quotes for the day, B_q (A_q) is the q*th* inside bid (ask) quote. The estimate in equation (5) is the daily variance of stock prices divided by average stock price for the day. The estimate is useful in reflecting information contained in quotes during the day. We take an average of the estimate of daily variance over each month, and use the monthly average for regression (4).

We estimate the regression (4) in two different ways. We run, for each month during sample period, a cross-sectional regression based on full sample, and then we run a regression based on entire stock-month panel data. Panels A and B of Table 3 report the results of regression based on full sample. We also estimate the regression (4) using samples classified by dollar trading volume. Panels A and B of Table 4 report the results of regression by dollar trading volume group.

Panel A of Table 3 summarizes a total of 68 cross-sectional regressions over our sample period. We report the cross-sectional mean of coefficients of 68 regressions for each variable. The number below the mean coefficient is the Wilcoxon signed rank test statistic and the p-value for the null hypothesis that the mean coefficient is equal to zero. Below the Wilcoxon test statistic are number of coefficients significantly negative and significantly positive at the 5% level (location count).

Panel A of Table 3 shows that the regression model (4) does a good job: adjusted R-squares are in the range of 0.39-0.70 with the mean R-square of 0.6041. Consistent with previous studies, we find that the percentage effective spreads are negatively related to trade size, and number of trades, and number of market makers and that spreads are positively related to the variance and inverse price. Note that the coefficients in Panel A are fairly close to the estimates based on stock-month panel data in Panel B. All coefficients are statistically significant at the 1% level based on the Wilcoxon test. The location count reveals that coefficients for HHI, variance and inverse price are always significantly positive, coefficients for trade size is always

significantly negative, and coefficients for number of trades are significantly negative in 64 months out of 68 months. Coefficients for the number of market makers are significantly negative about half the entire sample months.

What is interesting in Table 3 is that the HHI always has a significantly positive coefficient even after controlling for usual variables that explain trading costs. We already have included the number of market makers in the regression. The observation that HHI has a significant coefficient suggests that the way market shares are distributed among market makers affect trading costs. Or, the degree of market concentration affects trading costs so that the higher the concentration, the higher the trading costs.

The effect of concentration on trading costs is not trivial. The mean price, percentage effective spreads and effective spreads of entire sample are $17.92, 2.45% and ¢30.75. The mean coefficient on HHI is 0.257. Based on these numbers, a change of HHI from 2,000 to 1000 is associated with a decrease of percentage effective spread by 0.257% (0.257*(2,000-1,000)/1,000 = 0.257%). This corresponds to a decrease of effective spread by ¢4.61 ($17.92*0.257% = ¢4.61), or 15% of the mean effective spreads (¢4.61 ÷ ¢30.75 = 15%).

Table 4 provides a summary of regression by dollar trading volume group. It would be reasonable to expect that market makers in general face more stiff competition in making markets for high volume stocks than low volume stocks. Table 4 confirms this intuition. The coefficient on HHI for top 200 stocks (Group 3) is 0.09, only half of the coefficient for very low volume stocks (Group 1). Results in Table 4 are qualitatively similar to those in Table 3. All coefficients are statistically significant with correct sign. Adjusted R-squares are quite high. Results in Panel A and B are fairly consistent.

In sum, we provide evidence that concentration does affect trading costs even after controlling for usual explanatory variables. The effects are statistically significant. Point estimate implies non-trivial significance in economic sense. The remaining issue is to examine the economic significance of the effect of concentration. If they are indeed economically significant, an interesting issue is the longevity of the effect: will

the effect be persistent or will they quickly dissipate amid intense force of competition among market makers? We examine these issues next.

4. Economic Significance and Persistence of Rent

In previous section, we provide evidence that the degree of concentration affects trading costs. Our interpretation of the result is that market makers exercise market power in setting bid and ask prices, and they are able to extract rent. We examine in this section the economic significance and persistence of rent.

A. Economic Significance of Rent

We define our measure of rent as follows:

$$\text{Percentage Rent} = (\text{HHI}/1{,}000)*\text{Regression Coefficient} \quad (6)$$

$$\text{Rent} = \text{Average Closing Price} * \text{Percentage Rent}. \quad (7)$$

Percentage rent is the proportion of effective spread attributable to market power, expressed in percentage of closing price. For instance, percentage rent of 0.1% for a \$18 stock implies that effective spread includes the rent of ¢1.8 (\$18*0.1% = ¢1.8). For stock *i*, we estimate the percentage rent each month by multiplying the HHI_i with cross-sectional regression coefficient on HHI for that month. We use the product of percentage rent and average stock price as our estimate of rent (in cent) for stock *i* for that month. Once we estimate the percentage rent and rent (in cent) for all sample stocks over entire sample period, we calculate time-series mean of the percentage rent and rent (in cent). (See Figures 2-5 for time-series mean.) Figure 2 reveals a break point of the time series: up until February 1997, the rents move around ¢12, and thereafter rents have declined and fluctuated around ¢8. The decline of rent shown in Figure 2 is a result of the implementation of new order handling rules. (NASDAQ implemented first two phases of new order handling rules on January 20 and February 10, 1997.) Weston (2000) also observes the drop in economic rent following the

implementation of new order handling rules. Weston decomposes the bid-ask spreads using the approach of Huang and Stoll (1997). He estimates combined order processing cost and rent at ¢27 for 88 NASDAQ stocks before new rules, which declined by an average of ¢10 after new rules.

Figure 3 shows rents by dollar trading volume group. It appears that rent is about the same across dollar volume groups, except for a recent run-up for high volume stocks in early 2000. Since stock prices of high volume stocks tend to be higher than those of low volume stocks, the percentage rents for low volume stock should be higher than those of high volume stock. Figure 5 confirms this intuition: Percentage rent is higher for stocks in lower dollar volume group. We again see in Figure 5 a decline of percentage rent from early 1997, especially more so for stocks in dollar volume group 1. Figure 4 shows the mean percentage rent. Up until January 1997, percentage rent moved around 0.70%, while it goes down to around 0.50% since February 1997.

Figures 2-4 are based on time-series mean of rent (in cent) and percentage rent for stock *i* for entire sample period. We report in Table 5 cross-sectional average of those time-series mean of rent (in cent) and percentage rent across all sample stocks. Panel A is for full sample, Panel B is for dollar trading volume group, and Panel C is for group classified by year of initial public offering (IPO). Columns (1), (2) and (3) show the number of observations, rent in cent and percentage rent. (Disregard columns (4) – (6) for now.) For full sample, average rent is ¢8.76, and average percentage rent is 0.54%. Rent for each dollar trading volume group is about the same at about ¢9. Since stock price and dollar trading volume are positively correlated (see Table 2), and more actively trades stocks in general have higher prices, percentage rent is lower for actively traded stocks. The percentage rent is 0.22% for high trading volume group, 0.32% for medium volume group, and 0.58% for low volume group, which is 2.7 times higher than that for high volume group. The percentage rent for IPO group has declined from 0.59% for IPOs done in 1995 to 0.56% for 1996 and 0.50% for 1997. Average percentage rent for IPOs done prior to 1995 stand at 0.52%.

B. Persistence of Rent

Results in previous section suggest that market makers exercise market power in setting bid and ask prices, and that the magnitude of rent is non-trivial. It will be also important for both investors and regulators to know how quickly or slowly these rents dissipate. If they dissipate quickly via intense competition, then the presence of market power would be of less concern. However, if they are persistent, it is a different story. In this section, we examine the persistence of rent.

As our measure of persistence, we use half-life of percentage rent based on three-month moving average of percentage rent. Columns (4) and (5) show mean and standard deviation of half-life. For some stocks, their percentage rents have never dropped below one-half of initial level. Mean and standard deviation in columns (4) and (5) are only those stocks which have reached the level for half-life during sample period. Column (6) shows the percentage of those stocks out of entire sample stocks. For entire NASDAQ sample (Panel A), average half-life is 20.3 months for 79.4% of sample stocks. Panel B shows half-life for dollar trading volume groups: 23 months for high volume group, 20.6 months for medium volume group, and 20.2 months for low volume group. Half-life would be closely related with how stable the competitive environment for the market making activities are. For stocks in high and low trading volume groups, the competitive environment would be stable. The difference would be that high volume group stocks would face more stiff competition and low percentage rent, while low volume group stocks would face more concentration among a few dealers and high percentage rent. Consequently, we expect to see a shorter half-life for stocks in medium volume group than those for high and low volume groups, which is consistent with result in Panel B. Note that half-life of Group 1 stocks is 20.2 months, which appears to be shorter than that for Group 2 stocks. But the number is the average of only 77.4% of Group 1 stocks, which have reached the level for half-life during sample period.

Panel C shows half-life of groups classified by year of initial public offering. Since "old" stocks with IPOs prior to 1995 are likely to be at more mature stage in terms of competitive environment, those stocks would have longer half-life than "younger" stocks. Indeed, half-life of "old" stocks is 24.1 months, longer than average

of 20.3 months for entire sample. Half-lives of "younger" stocks with IPOs during 1995-1997 range from 12.6 - 17.8 months.

In sum, our results suggest that dominant market makers are able to extract rent in setting bid-ask spread. Our estimate of the rent is approximately 0.54% of stock price. Half-life of the percentage rent is approximately 20 months for entire sample, while half-life of younger stocks tend to be shorter than those of more mature stocks. Perhaps, it might take a while before a market maker can build up infrastructure to attract order flows, as Schultz suggests. In the meantime, existing market makers would be able to exercise market power and appear to earn abnormally large profits.

5. Conclusion

Recent studies suggest that trading costs are increasing in market concentration in market-making ((Ellis, Michaely and O'Hara (2002), Schultz (2002), Klock and McCormick (1999)). This finding implies that dealers set wider bid-ask spreads than they would in a competitive market, and dealers benefit from the wide spreads. Given the central role of competition among market makers in a dealer market, good understanding on the nature of market power, competition and trading costs
is of great importance to practitioners, researches and regulators.

Using extensive dealer trading volume data, we examine how the dynamics of market structure affect trading cost and aim to shed more light on the behavior of market makers in NASDAQ. Our dataset covers around 3,500 NASDAQ stocks over a period from May 1995 to December 2000. To the best of our knowledge, no studies have yet examined time-series pattern in market power and competition in NASDAQ based on such extensive data.

We find that dominant market makers are able to extract rent in setting bid-ask spread as much as 0.54% of stock price. We feel that this is non-trivial magnitude. The market power of dominant dealers appears to be persistent: Half-life of the percentage rent is approximately 20 months for entire sample. Our result is consistent with Schultz (2002) who argues that dealers might need to develop necessary infrastructure in order to capture a meaningful level of market share. He identifies access to order flow and informational advantage as two key elements of such

infrastructure. While entry and exit in NASDAQ market making business appears to be fluid with very low cost, it might take a while before a new entrant poses a significant competition to existing dominant market makers.

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Table 1. Summary statistics for Nasdaq National Market (NNM) stocks during a period from May 1995 to December 2000. The number of sample observations ranges from 3,180 to 3,623 each month.

Panel A: Daily $-trading volume (in $1,000)

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	3,290	20,630	23.1	92.1	358.5	1,374	10,882
1996	3,615	22,720	25.4	112.3	427.2	1,662	11,724
1997	4,804	41,818	31.0	137.0	463.8	1,756	13,763
1998	6,264	54,782	30.2	125.9	443.9	1,932	16,492
1999	12,344	92,901	28.6	117.0	486.8	2,975	34,067
2000	23,438	152,477	27.2	142.3	800.8	5,338	68,349
Full Sample	9,163	80,702	27.6	121.4	475.5	2,219	22,383

Panel B: Average Trade Size (in $)

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	27,382	27,645	5,194	10,543	18,667	35,076	77,341
1996	28,166	27,357	5,351	10,923	19,406	36,439	78,133
1997	26,727	25,731	4,885	10,149	18,676	34,812	73,940
1998	22,544	23,467	3,836	8,217	15,466	29,203	63,676
1999	17,073	17,486	3,184	6,433	11,504	21,747	48,612
2000	15,358	17,288	2,686	5,592	10,096	19,553	44,543
Full Sample	22,709	23,901	3,750	8,137	15,185	29,159	65,785

Panel C: Daily number of trades

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	71.8	262.4	2.4	6.7	18.5	53.0	258.5
1996	84.2	309.9	2.6	7.6	21.5	62.3	303.7
1997	108.4	531.7	3.1	9.8	25.8	71.5	338.4
1998	163.4	878.6	3.7	10.8	30.6	93.0	515.9
1999	355.2	1,571.9	4.3	13.9	46.0	181.5	1,312
2000	702.6	3,177.0	5.0	19.9	90.9	355.9	2,481
Full Sample	253.8	1,554.8	3.3	10.4	31.6	109.4	790.4

Panel D: Daily average closing price (in $)

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	16.65	13.48	3.23	7.37	13.44	21.84	41.14
1996	16.84	13.57	3.39	7.69	13.47	22.09	40.87
1997	17.14	14.27	3.13	7.14	13.53	22.94	42.95
1998	17.32	15.67	2.94	6.88	13.31	23.09	44.20
1999	18.27	21.12	2.93	6.60	12.17	22.13	53.71
2000	21.01	27.00	2.98	6.67	12.53	24.08	67.43
Full Sample	17.92	18.41	3.06	7.03	13.09	22.66	47.54

Panel E: Effective Spread (in $)

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	0.3822	0.265	0.137	0.208	0.330	0.464	0.838
1996	0.3781	0.279	0.126	0.199	0.312	0.463	0.854
1997	0.3220	0.249	0.098	0.168	0.256	0.394	0.758
1998	0.2927	0.268	0.082	0.141	0.218	0.351	0.743
1999	0.2405	0.224	0.076	0.125	0.185	0.284	0.569
2000	0.2516	0.293	0.078	0.127	0.188	0.293	0.604
Full Sample	0.3075	0.269	0.088	0.155	0.236	0.380	0.735

Panel F: % Effective Spread

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	3.19	2.36	0.75	1.62	2.58	4.06	7.75
1996	2.98	2.13	0.75	1.53	2.45	3.81	7.00
1997	2.45	1.82	0.61	1.29	2.03	3.07	5.69
1998	2.30	1.75	0.52	1.11	1.87	2.98	5.56
1999	2.03	1.74	0.42	0.94	1.60	2.65	5.04
2000	2.00	1.67	0.39	0.88	1.55	2.61	5.11
Full Sample	2.45	1.94	0.52	1.16	1.97	3.16	6.01

Panel G: Number of active market makers

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	11.6	7.2	3	7	10	15	25
1996	10.6	6.9	3	6	9	14	24
1997	11.7	7.4	3	6	10	15	26
1998	13.3	8.3	4	7	12	17	29
1999	15.1	8.9	4	9	14	19	33
2000	16.5	8.9	5	10	15	21	34
Full Sample	13.2	8.3	4	7	11	17	29

Panel H: Herfindhal Index

			Percentile				
	mean	std	5%	25%	50%	75%	95%
1995	2,261	1489	701	1,221	1,834	2,869	5,274
1996	2,342	1539	697	1,228	1,913	3,021	5,464
1997	2,268	1468	718	1,227	1,859	2,869	5,225
1998	2,214	1432	708	1,209	1,806	2,804	5,078
1999	1,983	1270	650	1,130	1,638	2,449	4,522
2000	1,748	1118	583	1,004	1,448	2,155	3,894
Full Sample	2,133	1,406	669	1,160	1,734	2,683	4,980

Table 2. Correlation among key variables. Data are for Nasdaq National Market stocks during a period from May 1995 to December 2000.

	$-volume	TSize	#Trade	% effect sp	HHI	# mm	Cap	Price
$-volume	1	0.14	0.9	-0.12	-0.11	0.33	0.82	0.36
Trade Size	0.14	1	0.09	-0.34	0.11	0.09	0.12	0.53
# of trade	0.9	0.09	1	-0.15	-0.15	0.41	0.67	0.31
% effect sp	-0.12	-0.34	-0.15	1	0.37	-0.47	-0.08	-0.39
HHI	-0.11	0.11	-0.15	0.37	1	-0.62	-0.08	-0.13
# of mm	0.33	0.09	0.41	-0.47	-0.62	1	0.24	0.24
Cap	0.82	0.12	0.67	-0.08	-0.08	0.24	1	0.23
Price	0.36	0.53	0.31	-0.39	-0.13	0.24	0.23	1

Table 3. Coefficients of regression of percentage effective spread on Herfindahal index, and other control variables: % Spread i = alpha0 + alpha1 HHIi + alpha2 ln(Tsize i) + alpha3 ln(Ntrades i) + alpha4 sigma*sigma i + alpha5 [1/E(Pi)] +alpha6 Ni + e i. All variables are monthly averages. % Spread i denotes percentage effective spread; HHI, Herfindahl index; Tsize, size of trade in dollar; Ntrades, number of daily trades; sigma*sigma, a measure of variance of stock return; E(Pi), average stock prices; Ni, number of market makers for stock i. The regressions are based on data on Nasdaq National Market (NNM) stocks from May 1995 to December 2000. Panel A reports means of cross-sectional regression each month. The numbers below the coefficient means are the Wilcoxon statistics for a null hypothesis that the mean is equal to 0, and the p-value. Below the Wilcoxon statistics are number of coefficients significantly negative and significantly positive at the 5% level.

	Panel A Cross-sectional Regression	Panel B Panel Data Regression
Intercept	4.3414 (1173, 0.0001) (0, 68)	3.8919 (211.02)
Herfindahl Index/1000	0.257 (1173, 0.0001) (0, 68)	0.2308 (87.15)
Ln (Trade size)	-0.6563 (-1173, 0.0001) (68, 0)	-0.4493 (-93.93)
Ln (Number of Trades)	-0.3229 (-1173, 0.0001) (64, 0)	-0.3192 (-94.65)
Variance	1.0025 (1173, 0.0001) (0, 68)	0.2435 (41.76)
1/Price	4.6543 (1173, 0.0001) (0, 68)	5.9577 (138.95)
Number of market makers	-0.016 (- 552, 0.0005) (32, 18)	-0.0210 (-32.03)
Adjusted R-square	0.6041	0.5173

Table 4. Coefficients of regression of percentage effective spread on Herfindahal index, and other control variables. The regressions are based on data on Nasdaq National Market (NNM) stocks from May 1995 to December 2000. Cross-sectional regressions are done by $-trading volume group each month. we classify top 200 stocks in terms of $-trading volume to $-volume group 1, next 800 stocks to group 2 and the remaining stocks to group 3. The first number is time-series mean of coefficients from cross-sectional regression each month. The numbers below the coefficient means are the Wilcoxon statistics for a null hypothesis that the mean is equal to 0, and the p-value. Below the Wilcoxon statistics are number of coefficients significantly negative and significantly positive at the 5% level. Adjusted R-square are the time series means of adjusted R-square of each cross-sectional regression.

Panel A. Cross-sectional Regression by $-trading volume group

	$-trading volume		
	Group 3	Group 2	Group 1
Intercept	1.488	3.2918	5.7548
	(1173, 0.0001)	(1173, 0.0001)	(1173, 0.0001)
	(0, 67)	(0, 68)	(0, 68)
Herfindahl Index/1000	0.0953	0.1362	0.1878
	(1118, 0.0001)	(1173, 0.0001)	(1173, 0.0001)
	(1, 47)	(0, 65)	(0, 67)
Ln (Trade size)	-0.1622	-0.3882	-0.8827
	(-1173, 0.0001)	(-1173, 0.0001)	(-1173, 0.0001)
	(66, 0)	(66, 0)	(68, 0)
Ln (Number of Trades)	-0.0639	-0.2127	-0.4019
	(-1159, 0.0001)	(-1173, 0.0001)	(-1165, 0.0001)
	(59, 0)	(65, 0)	(64, 0)
Variance	0.4252	0.6449	1.3491
	(1173, 0.0001)	(1173, 0.0001)	(1173, 0.0001)
	(0, 65)	(0, 67)	(0, 67)
1/Price	1.9861	1.4345	4.7744
	(892, 0.0001)	(732, 0.0001)	(1173, 0.0001)
	(9, 37)	(13, 37)	(0, 68)
Number of market mak	-0.004	-0.0135	-0.0762
	(-1064, 0.0001)	(- 1146, 0.0001)	(- 1169, 0.0001)
	(38, 0)	(55, 1)	(64, 0)
Adjusted R-square	0.7209	0.5791	0.5495

Table 4 (Continued)

Panel B. Panel Data Regression by $-trading volume group

	$-trading volume		
	Group 3	Group 2	Group 1
Intercept	1.6301	3.7555	5.2809
	(48.67)	(124.08)	(203.96)
Herfindahl Index/1000	0.0684	0.1162	0.1469
	(12.44)	(39.38)	(43.63)
Ln (Trade size)	-0.1224	-0.3722	-0.6230
	(-21.77)	(-69.36)	(-91.29)
Ln (Number of Trades)	-0.0795	-0.2728	-0.4017
	(-26.57)	(-74.88)	(-75.97)
Variance	0.1848	0.3245	0.2611
	(29.63)	(57.54)	(36.42)
1/Price	3.4542	2.4909	6.5799
	(26.59)	(38.69)	(126.23)
Number of market makers	-0.0088	-0.0215	-0.0865
	(-27.05)	(-47.6)	(-69.6)
Adjusted R-square	0.5130	0.4347	0.4396

Table 5. Estimation of rent and half-life of % rent. Each month, we run a cross-sectional regression of % effective spread on the Herfindahl index (HHI) and several other control variables. For each stock and each month, we use the product of regression coefficient on HHI and stock i's HHI as our estimate of % rent (i.e., rent as a % of stock price) for dealers making a market for stock i. We multiply this % rent by average closing price of stock i for that month and obtain our estimate of rent in cents. We calculate the time-series means of rent in cent and % rent for stock i, and then we take cross-sectional average the two variables across all stocks. They are in column (2) and (3). For calculation of half-life, we use a three-month moving average of % rent. The half-life of stock i is the time until the observatio of % rent for stock i halves. Column (4) reports cross-sectional average of half-life for stock i, Column (5) reports standard deviation. Column (6) reports the percentage of stocks which have reached the level of % rent for half-life during sample period.

				Half-life in month		
	(1)	(2)	(3)	(4)	(5)	(6)
	Nobs	rent in cents	rent in % price	mean	st dev	% of (1)
Panel A. Full sample						
	3,580	8.76	0.5434	20.3	10.4	79.4
Panel B. By $-trading volume group						
Group 3	106	9.83	0.2193	23.0	11.8	88.7
Group 2	397	9.23	0.3213	20.6	9.8	92.2
Group 1	3,077	8.67	0.5832	20.2	10.4	77.4
Panel C. By year of initial public offering						
prior to 1995	1,860	8.96	0.5202	24.1	10.6	82.5
1995	285	9.89	0.5939	17.8	8.0	90.9
1996	346	8.61	0.5606	12.6	5.7	93.4
1997	119	8.19	0.5043	12.8	8.6	77.3

Figure 1: Herfindahl Index

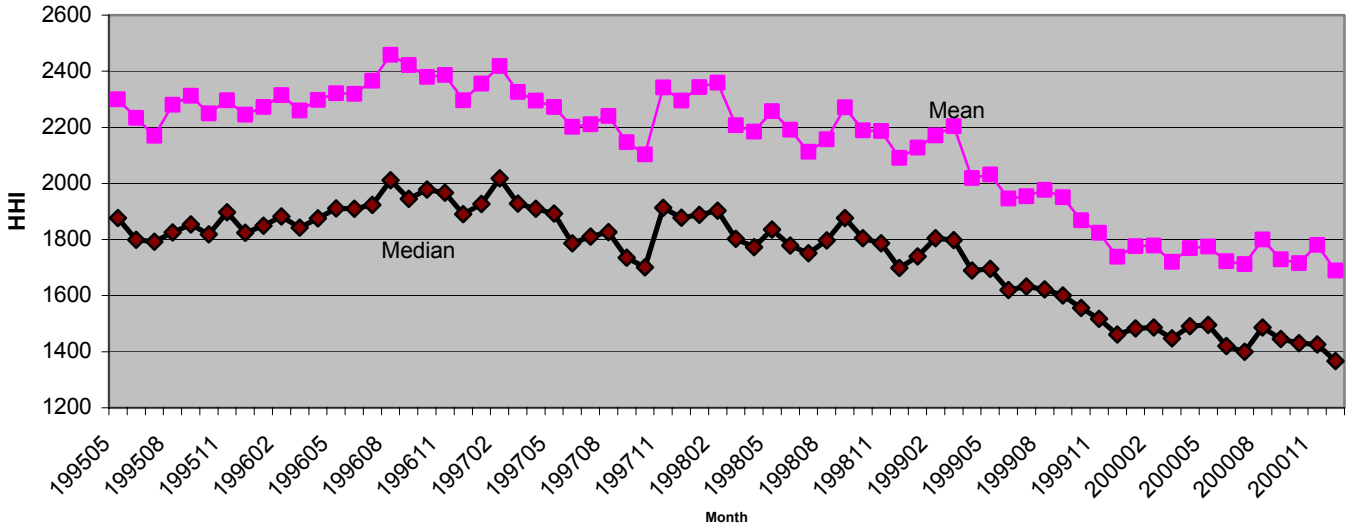


Figure 1B: HHI by $-trading volume group

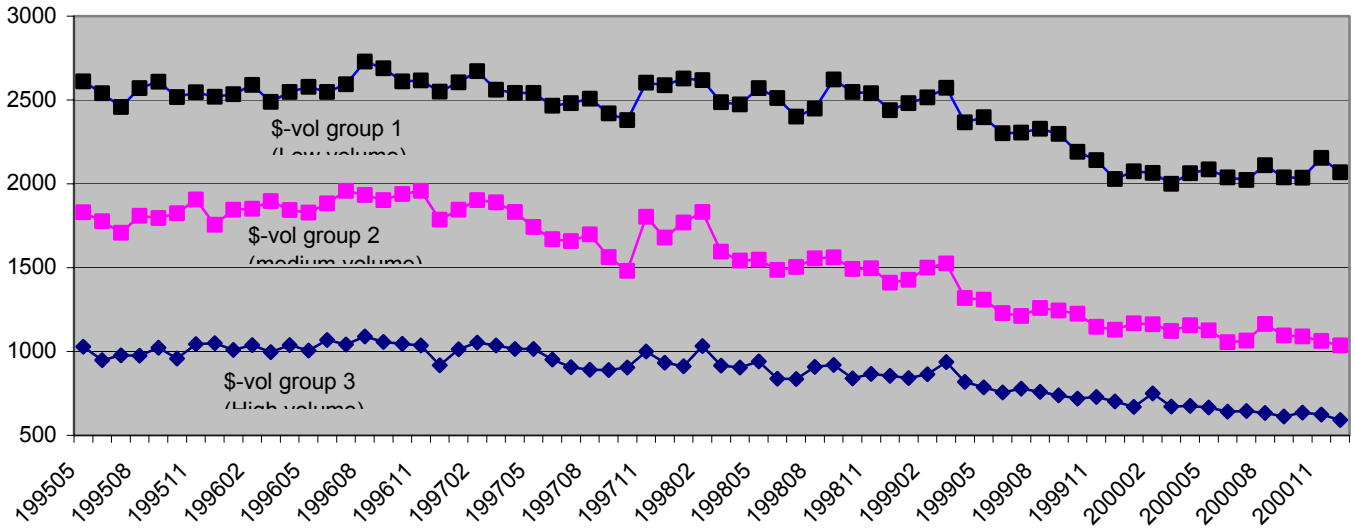


Figure 2: Rent in cents

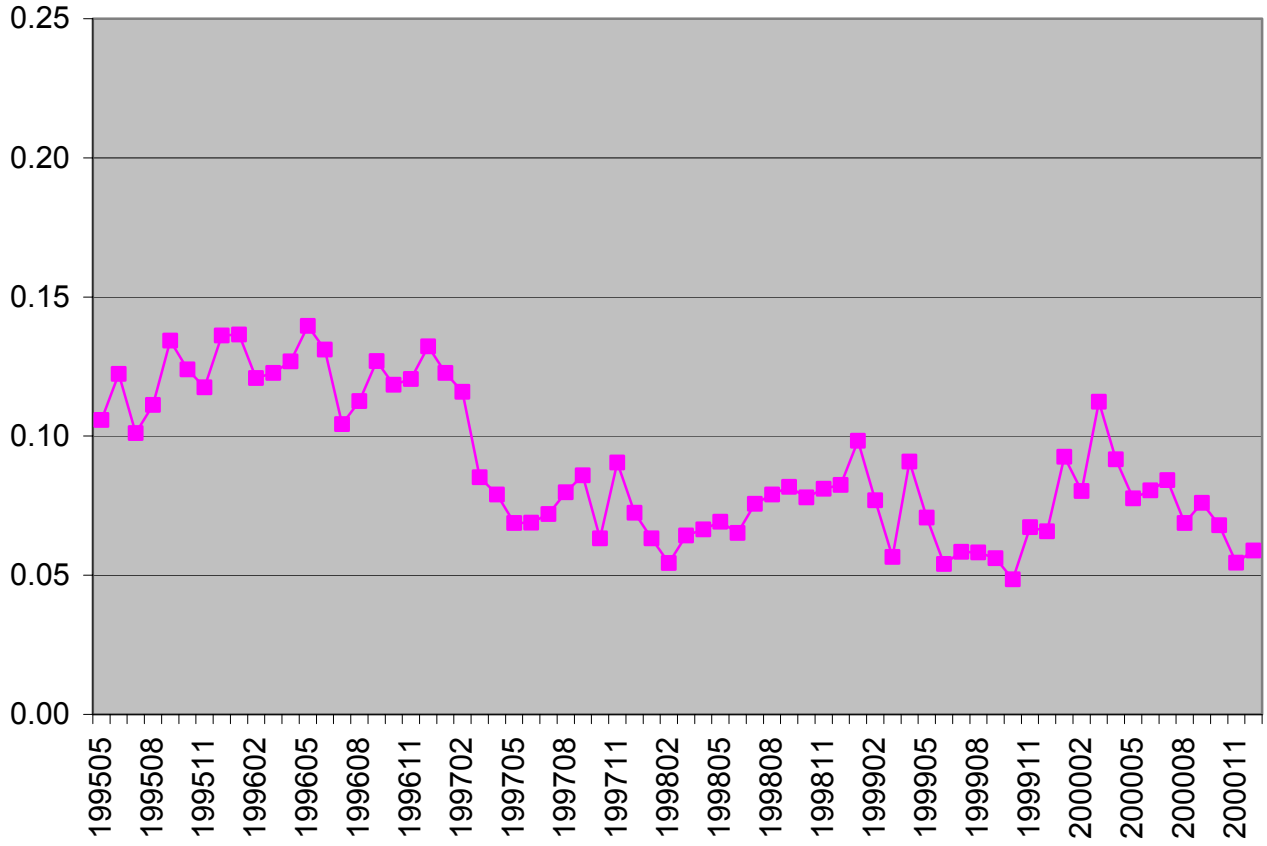


Figure 3: Rent in cents by $-volume group

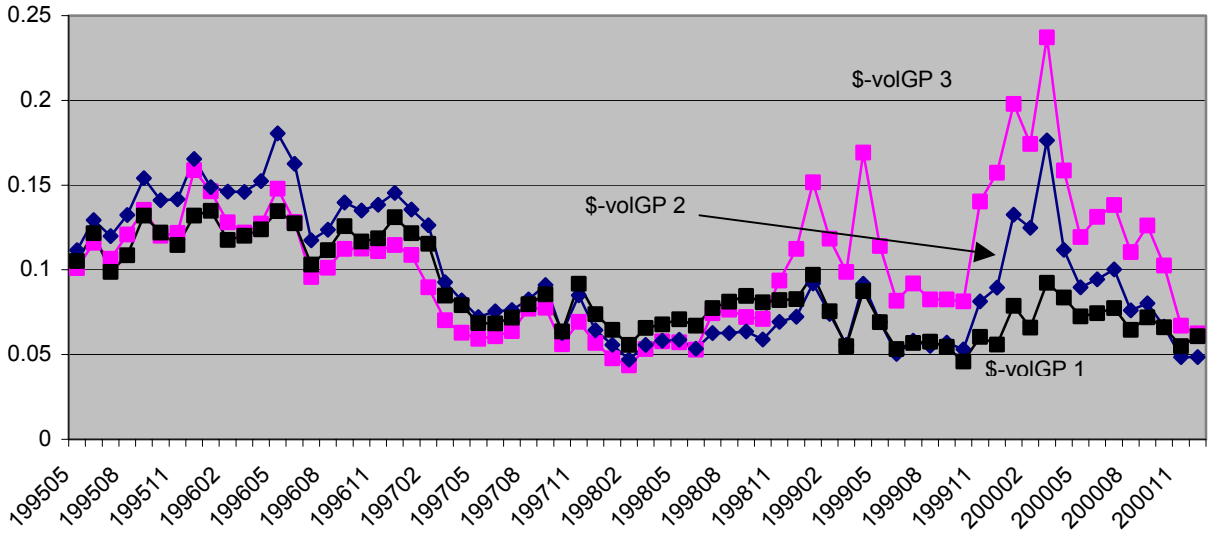


Figure 4: Rent in % of Stock Price

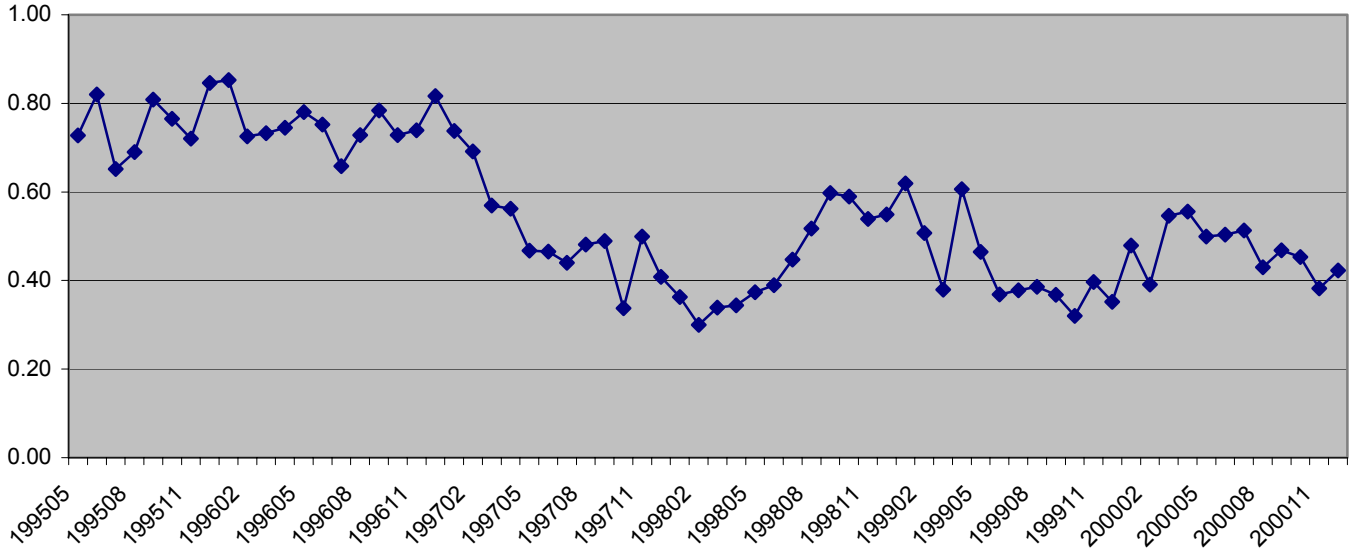


Figure 5: Rent in % by $-volume group

